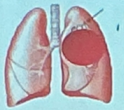
Once Cushing;s is confirmed:

What is the Cause?= Differential Diagnosis



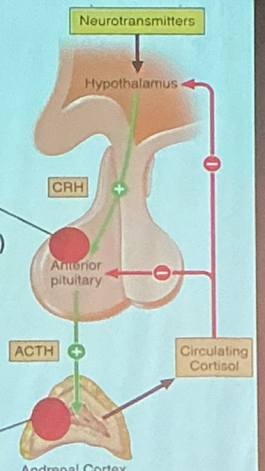
Ectopic ACTH source

ACTH-dependent Cushing's

Pituitary ACTH-producing tumour (ben.) **Cushing's disease**

ACTH-independent (=adrenal) Cushing's

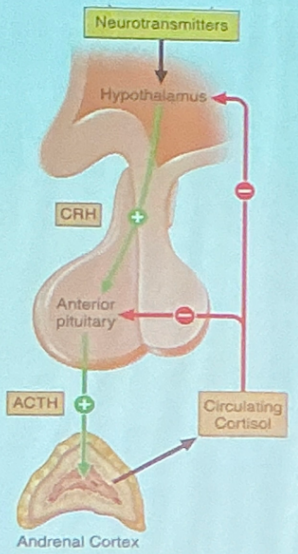
Adrenal cortisol-producing tumour (benign or malignant)



Cushing is confirmed:

What is the cause? = Differential Diagnosis

- 09.00h Plasma ACTH
  - ACTH suppressed => Adrenal
  - ACTH normal/high => Pituitary or Ectopic
- Pituitary or Ectopic?
  - High Dose Dex Test (2mg 6hourly for 48 h)
  - CRH test (100µg i.v.)
  - Inferior petrosal sinus sampling (IPSS)
- Only then – Imaging (CT adrenals/MRI pituitary)



Biochemistry before Imaging

Adrenal cortisol-producing tumor

hyperpigment → ACTH-dependent

ไม่มี → independent.

Treatment of Cushing's Syndrome

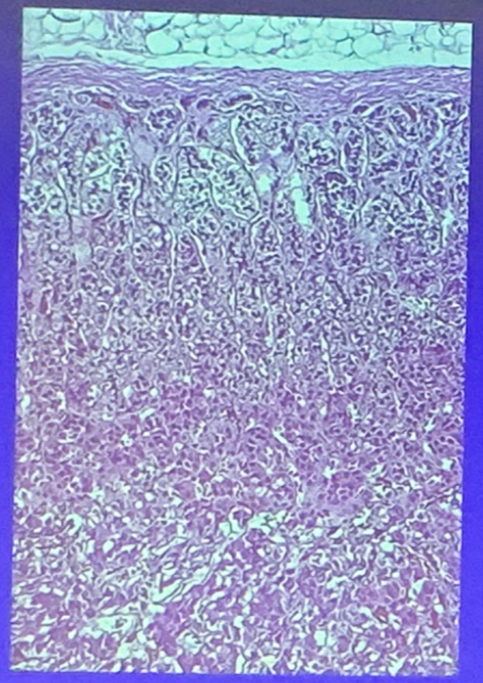
- **Surgery**
  - Pituitary (Cushing's disease) - Transsphenoidal surgery (hyperpigmentation)
  - bilateral adrenalectomy (=> cave Nelson's syndrome) → replace hormone
  - Laparascopic adrenalectomy for adrenal adenoma
  - Open adrenalectomy for adrenocortical carcinoma
- **Drugs**
  - block cortisol-producing adrenal enzymes (metyrapone, ketoconazol, etomidate) osilodrostat
  - block the glucocorticoid receptor (RU486)
  - disrupt adrenal redox balance and thereby steroidogenesis and cell proliferation (mitotane) analytic

bilateral adrenalectomy

→ Nelson's syndrome.

Adrenal Dysfunction

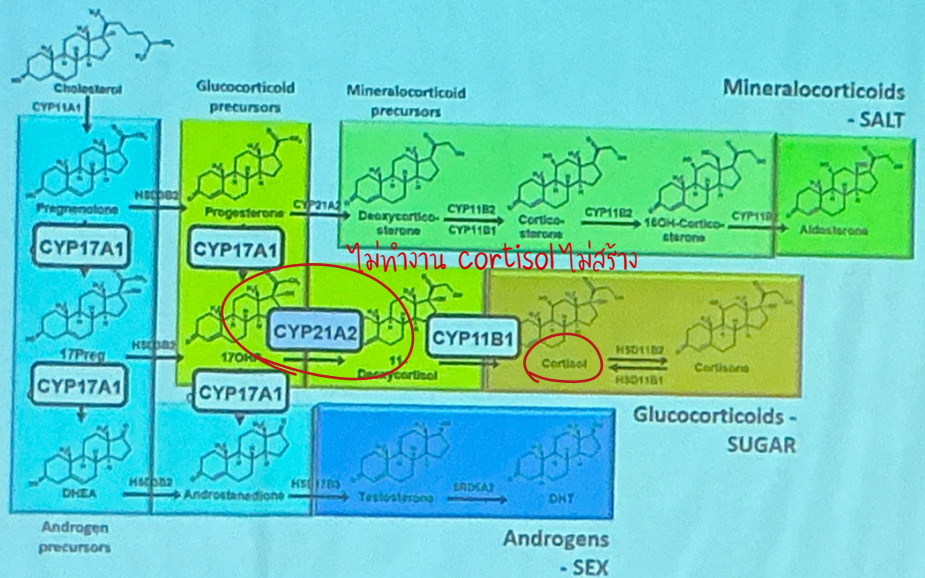
Adrenal Hormone Production



Zona Glomerulosa	Aldosterone „*Salt*“
Zona Fasciculata	Cortisol „*Sugar*“
Zona Reticularis	DHEA „*Sex*“
Adrenal Medulla	Catecholamines Epinephrine = Adrenaline → peptide

*

Congenital adrenal hyperplasia – Block in Adrenal Steroidogenesis affecting glucocorticoid synthesis

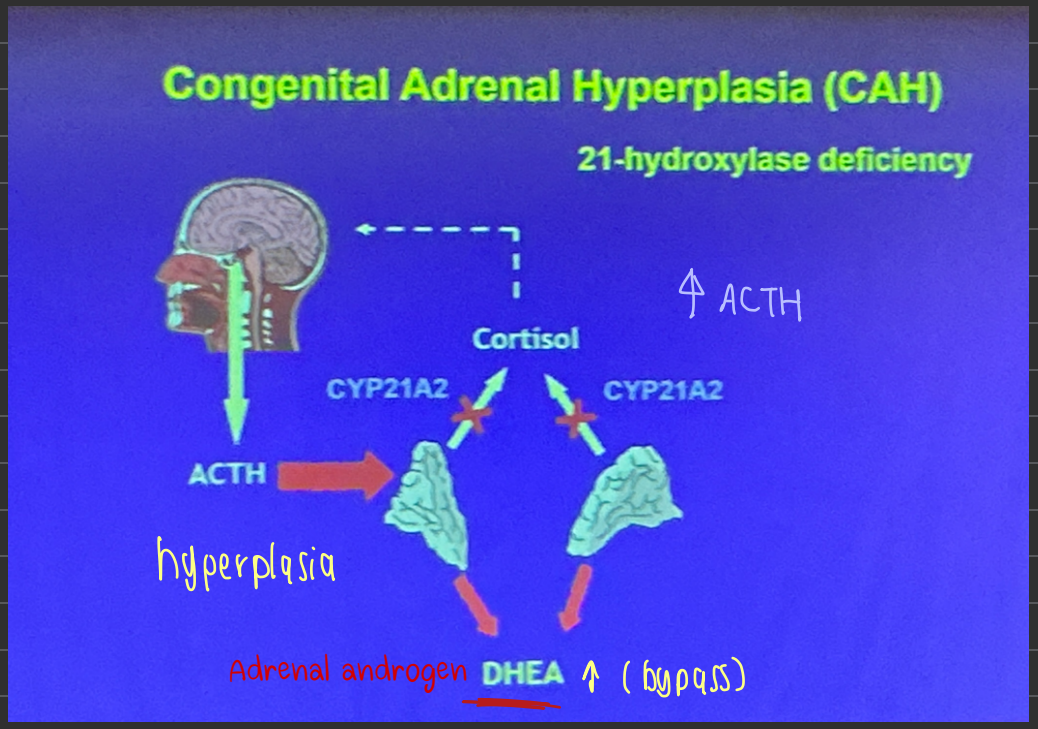


CYP21A2 def.

AI insuffiency

Bilateral adrenal hyperplasia

Aldosterone-producing adenoma?



21A2 def.

⇕

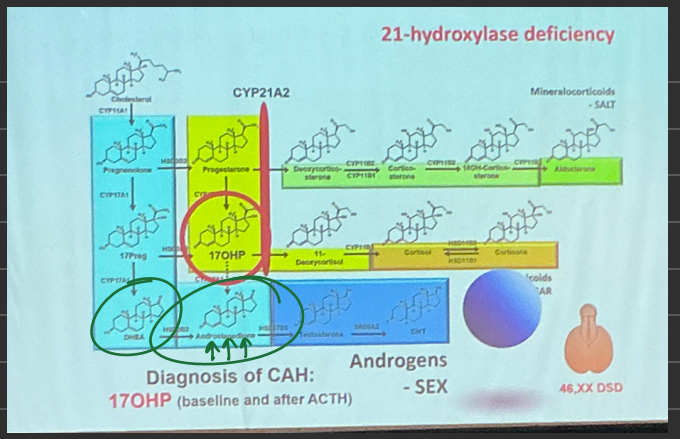
glucocorticoid

shift pathway Androgen*

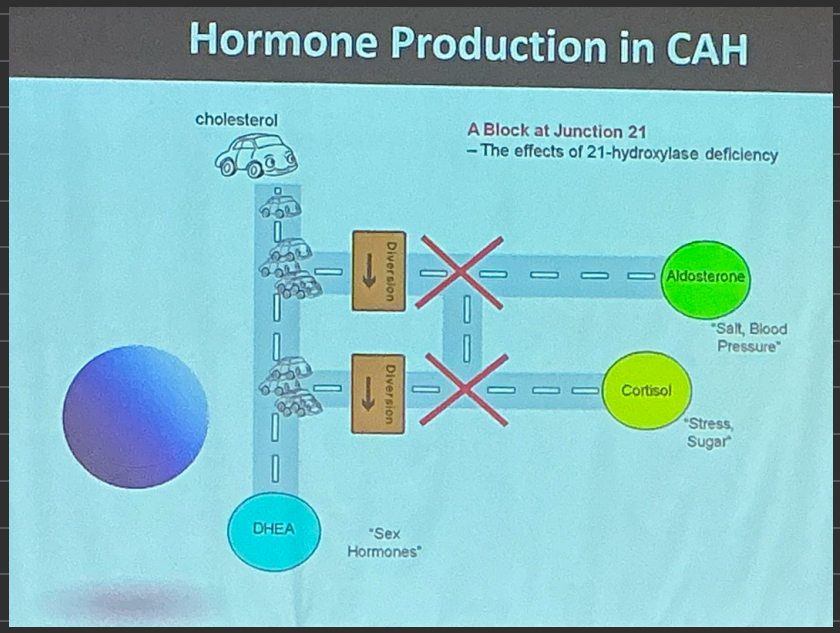
46,

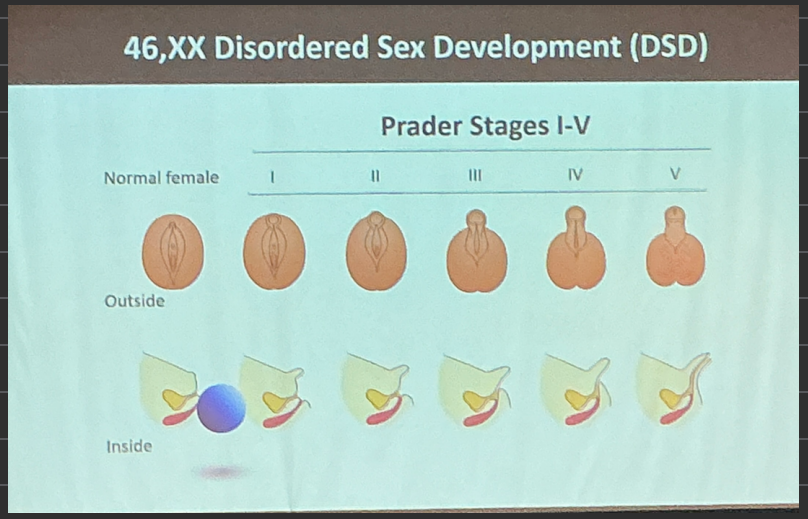
* 17 OHP (baseline & After ACTH)

CAH → 17 OHP ↑↑



genetalia abnormal





CAH

Classic CAH

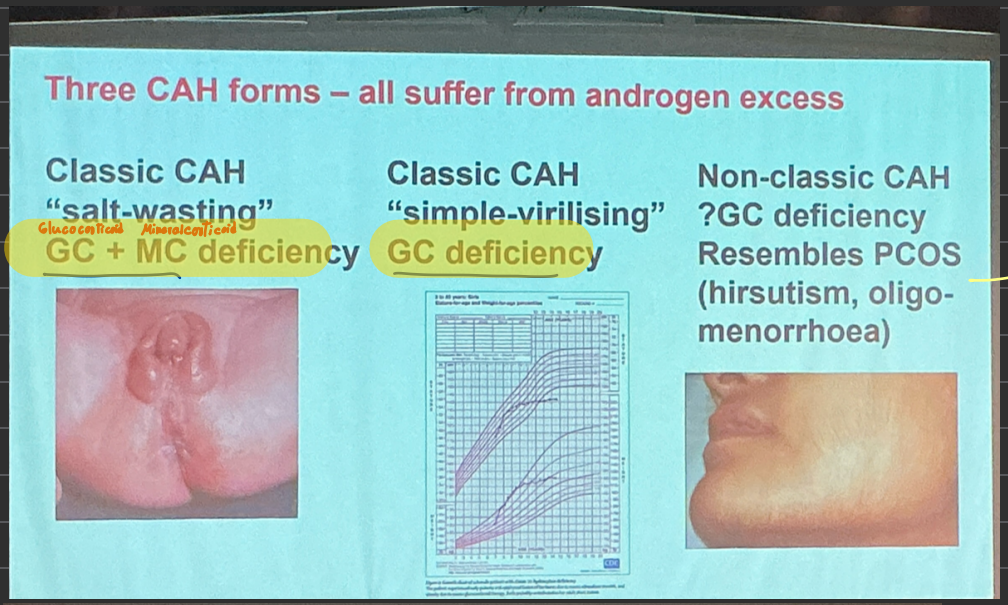
GC+ MC def.

hypernatremia, hypokalemia

Androgen excess

CAH

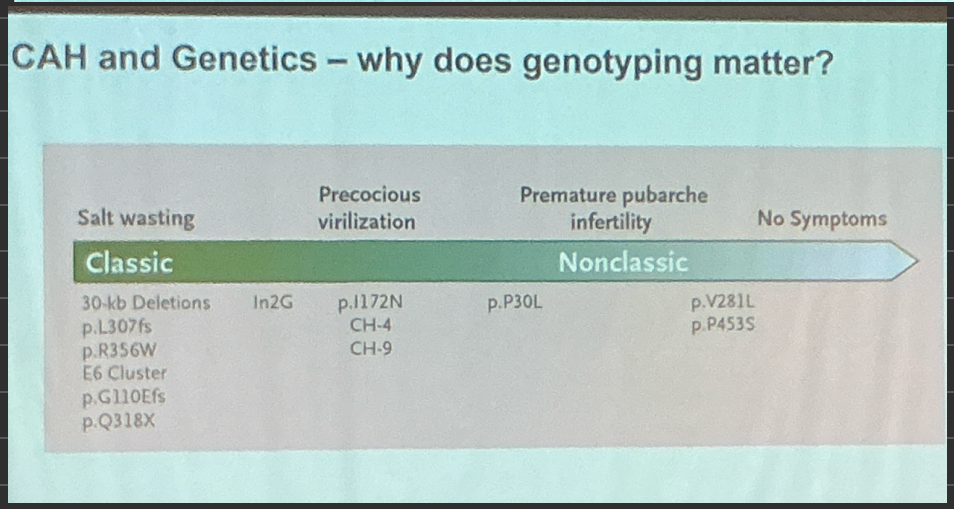
CAH
- salt-wasting
- simple virilising (masculinization)



แยกออกจาก PCOS

DX by exclusion

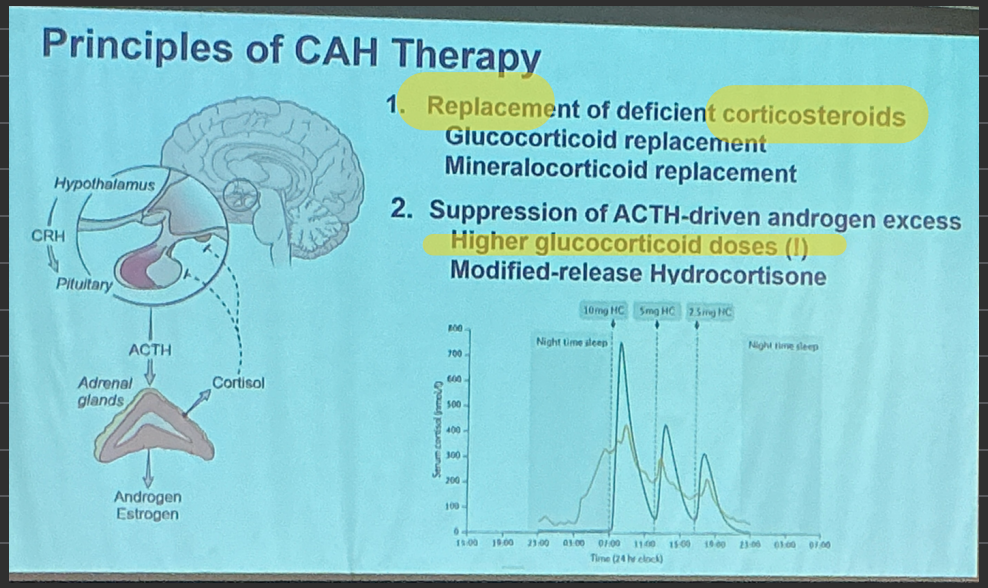
R/O CAH

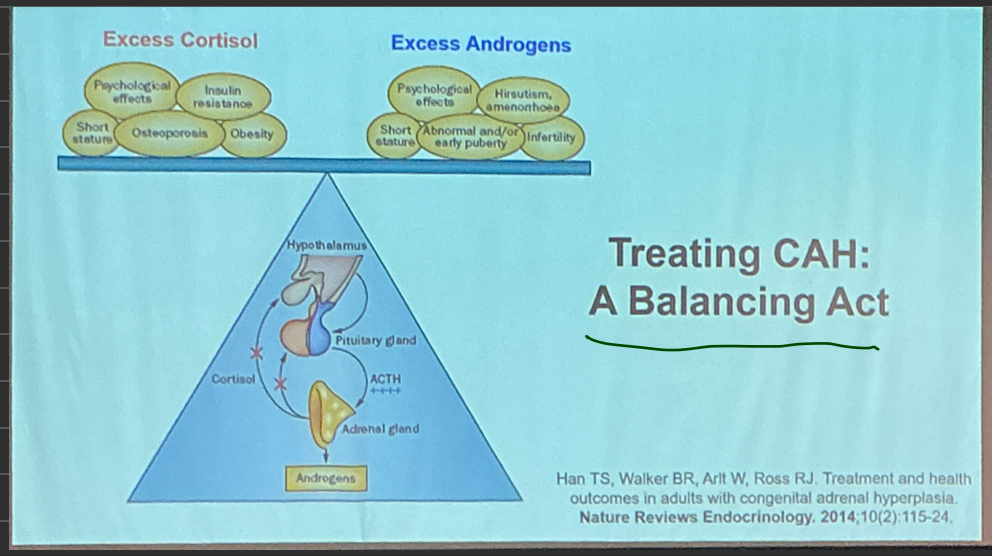


CAH Therapy *

↓↓ ACTH 용 glucocorticoid high dose

Modified-release hydrocortisone





↑ glucocorticoid excess ⊣ FH, LH

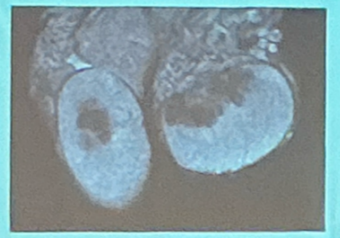
→ oligo / Amenorrhea

Fertility in women with CAH and men with CAH

Too much glucocorticoid

=> Suppression of LH and FSH

=> Oligo-/Amenorrhea



Not enough glucocorticoids

=> Androgen and progesterone excess

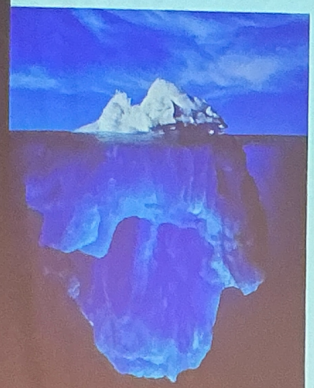
=> Anovulation, oligomenorrhoea, failure to implant

Testicular adrenal rest tissue (TART)

Arises from urogenital ridge, adrenal and gonadal features

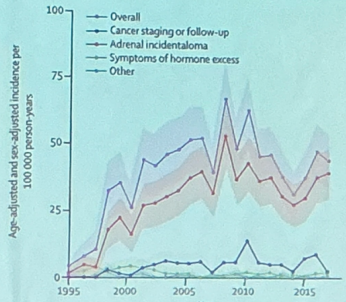
In particular in patients with poor control, impairs fertility

Adrenal Incidentalomas - The Scale of the Problem



- Adrenal tumours are discovered on 5% of cross-sectional imaging procedures
- In 2014, 81 Million CT scans in the United States (3M in 1980)

– the Adrenal Incidentaloma Iceberg.....

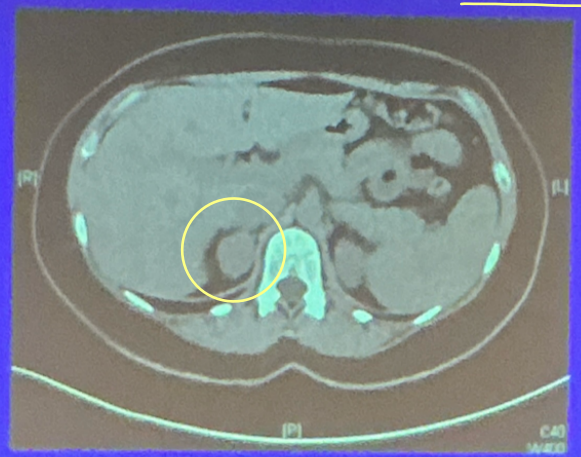


Epidemiology of adrenal tumours in Olmsted County, Minnesota, USA: a population-based cohort study.
Ebbehoj A, Li D, Kaur RJ, Zhang C, Singh S, Li T, Atkinson E, Achenbach S, Khosla S, Arlt W, Young WF, Rocca WA, Bancos I.
Lancet Diabetes Endocrinol. 2020 Nov; 8(11):894-902.

199 Incidentoloma ↑↑↑

Geraldine, 51 yrs

- August 2012: imaging of the abdomen for unspecific lower abdominal pain => 2.8 cm mass in the right adrenal gland



Endocrine Work-up in Adrenal Tumours

Exclude aldosterone excess (primary aldosteronism) ✓

- BP ↑ or K ↓: Paired Plasma Renin and Plasma Aldosterone

Exclude cortisol excess

- Serum cortisol at 8-9 a.m. after 1mg dexamethasone at 11 pm = overnight dexamethasone suppression test (DST) ✓

Exclude adrenal androgen excess

- Serum DHEAS, androstenedione, 17OHP (in tumours >4cm)

Exclude Phaeochromocytoma

- Plasma metanephrines ✓

Endocrine Work-up in Adrenal Tumours

Exclude aldosterone excess (primary aldosteronism)

- BP ↑ or K ↓: Paired Plasma Renin and Plasma Aldosterone

Exclude cortisol excess

- Serum cortisol at 8-9 a.m. after 1mg dexamethasone at 11 pm = overnight dexamethasone suppression test (DST)

Low dose (1 mg)

Exclude adrenal androgen excess

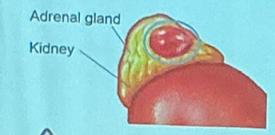
- Serum DHEAS, androstenedione, 17OHP (in tumours >4cm)

Exclude Phaeochromocytoma

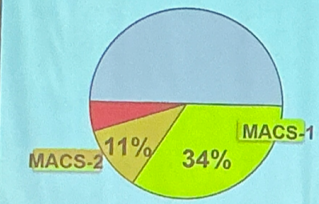
- Plasma metanephrines

↑ BP, Panic attack, Afib.

Mild autonomous cortisol secretion (MACS)



EURINE-ACT N=1,305



MACS is associated with an **increased risk of cardio-metabolic comorbidities:**

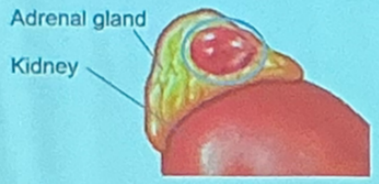
MACS-1 **5.0-fold** risk
MACS-2 **6.7-fold** risk
Hypertension

MACS-1 **2.8-fold** risk
MACS-2 **4.1-fold** risk
Type 2 diabetes

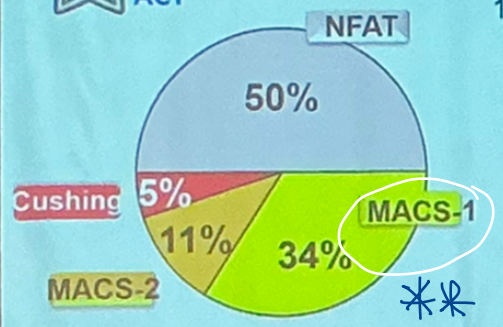
MACS-1 **2.2-fold** risk
MACS-2 **2.7-fold** risk
Dyslipidaemia

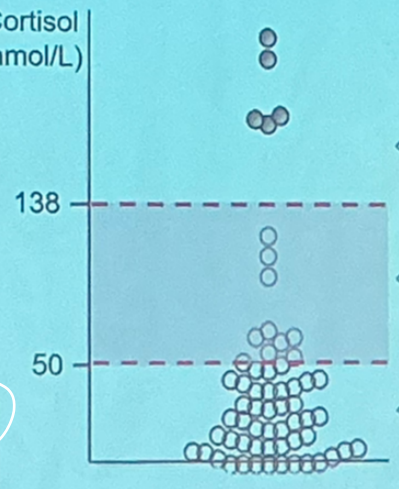
Prete A et al., Ann Intern Med 2022

Mild autonomous cortisol secretion (MACS)



EURINE-ACT N=1,305





1mg-dexamethasone suppression test (1mgDST)

>138 nmol/L: MACS-2
Autonomous cortisol secretion "confirmed"

51-138 nmol/L: MACS-1
Autonomous cortisol secretion "possible"

≤50 nmol/L: Normal
Autonomous cortisol secretion "excluded" (NFAT)

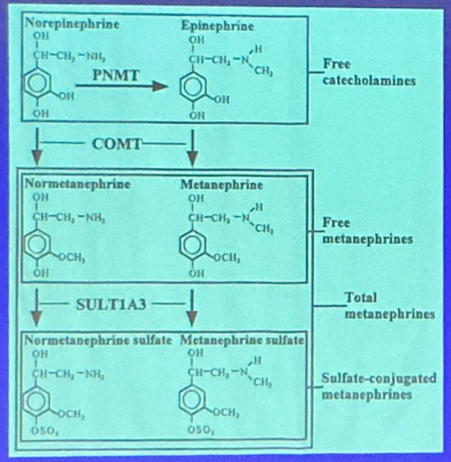
Prete A et al., Ann Intern Med 2022

Pheochromocytoma ตรวจ Plasma metanephriner sensitive

24 h - urinary meta 4 sp. *

→ Malignancy ?

Phaeochromocytoma Diagnosis

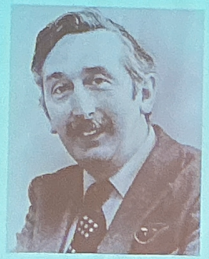


- Most sensitive: ☆ 24-h urine / serum * ☆ metabolize
- Plasma metanephrines
- Slightly less sensitive, but slightly more specific:
- 24-h urinary metanephrines

Tumour Radiodensity or Tumour Attenuation Value (measured in Hounsfield Units)

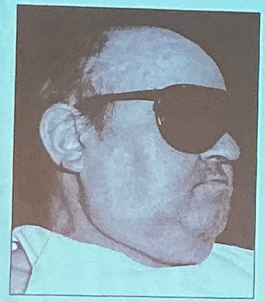
The **Hounsfield unit (HU) is a relative quantitative measurement of radiodensity** used by radiologists in the interpretation of computed tomography (CT) images.
The absorption/attenuation coefficient of radiation within a tissue is used during CT reconstruction to produce a grayscale image.
0 HU = density of distilled water

The Nobel Prize in Physiology or Medicine 1979 "for the development of computer-assisted tomography" (to Geoffrey Hounsfield and Alan McCormack)



Geoffrey Hounsfield

Adrenal Incidentalomas - The Scale of the Problem



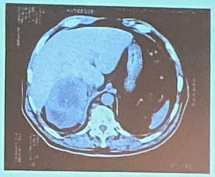
Woman with virilizing ACC 1987

- **Adrenocortical carcinoma (ACC)**, a highly malignant tumor with a poor prognosis, is rare but accounts for 5% of adrenal masses undergoing work-up
- Differentiating **adrenocortical adenomas (ACA)** from **ACC** represents a continuous challenge: **Imaging is the mainstay of differential diagnosis BUT provides poor sensitivity and specificity**

Agressive มากๆ

Adrenocortical carcinoma.

Tumor onset เร็ว รุนแรง
CAH เป็นเรื้อรังๆ ไม่รุนแรง



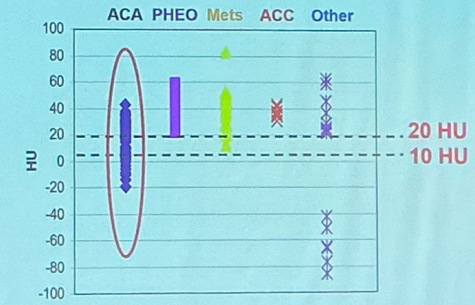
Adenoma vs. Carcinoma

Adenoma : homogenous HU ต่ำ

carcinoma : heterogenous HU สูง

Tumour Radiodensity in Hounsfield Units (HU) on CT
in 156 patients with histologically defined adrenal tumours

Adrenocortical Adenoma (ACA)



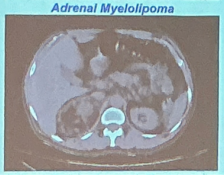
Hamrahian A et al., J Clin Endocrinol Metab 2005; 90(2):871-877.

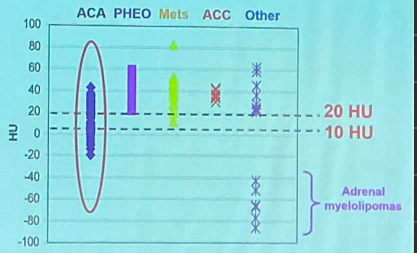
✓
✓
sure HU < 10 → R/O Pheo

Tumour Radiodensity in Hounsfield Units (HU) on CT
in 156 patients with histologically defined adrenal tumours

Adrenocortical Adenoma (ACA)

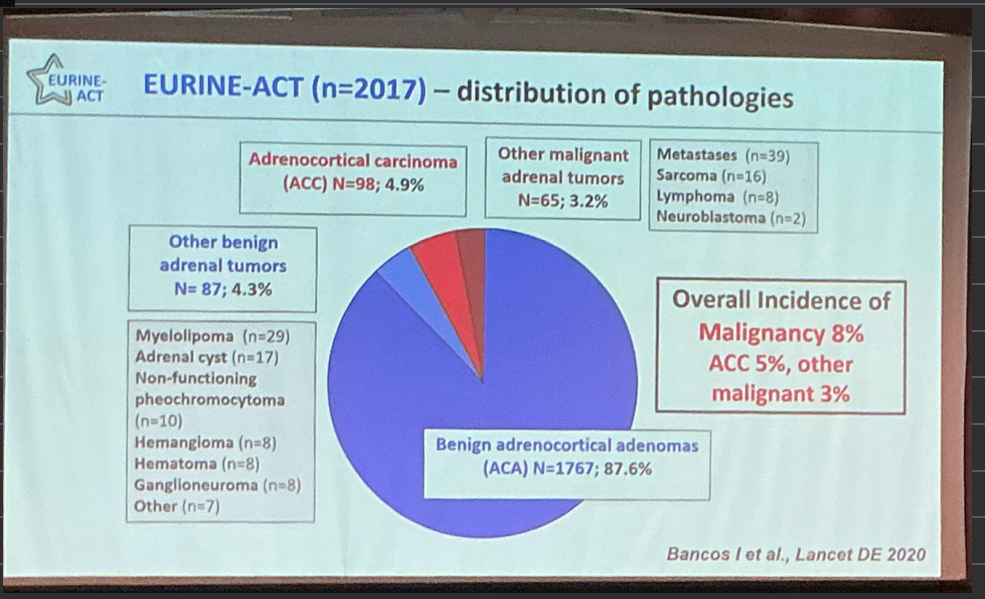
Adrenal Myelolipoma





Hamrahian A et al., J Clin Endocrinol Metab 2005; 90(2):871-877.

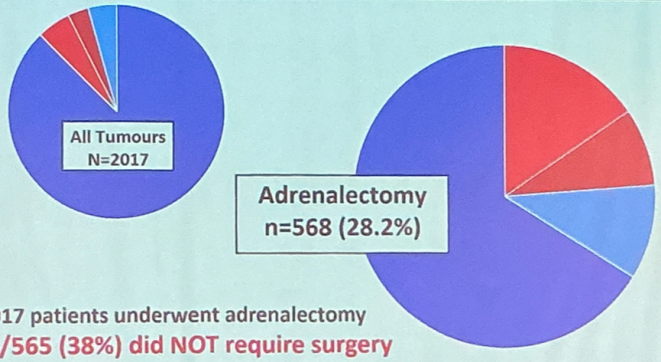
EURINE-ACT (n=2017) – distribution of pathologies



Bancos I et al., Lancet DE 2020

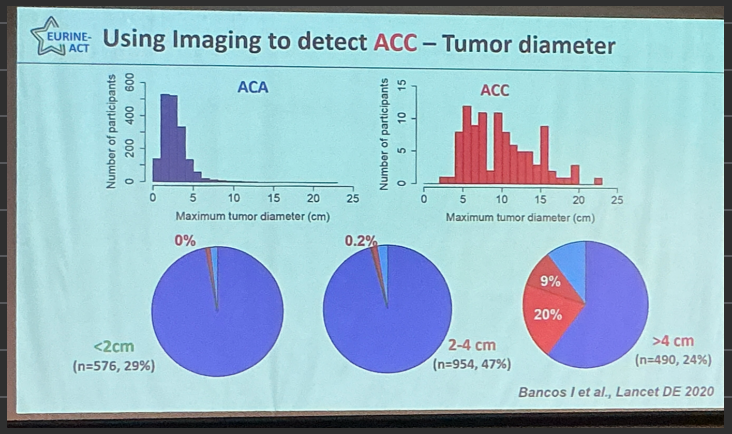
Adrenal tumor
↳ benign / non-function.

Adrenalectomy in the EURINE-ACT Cohort



568/2017 patients underwent adrenalectomy
BUT 214/565 (38%) did NOT require surgery
as they had a benign, non-functional mass

Bancos I et al., Lancet DE 20

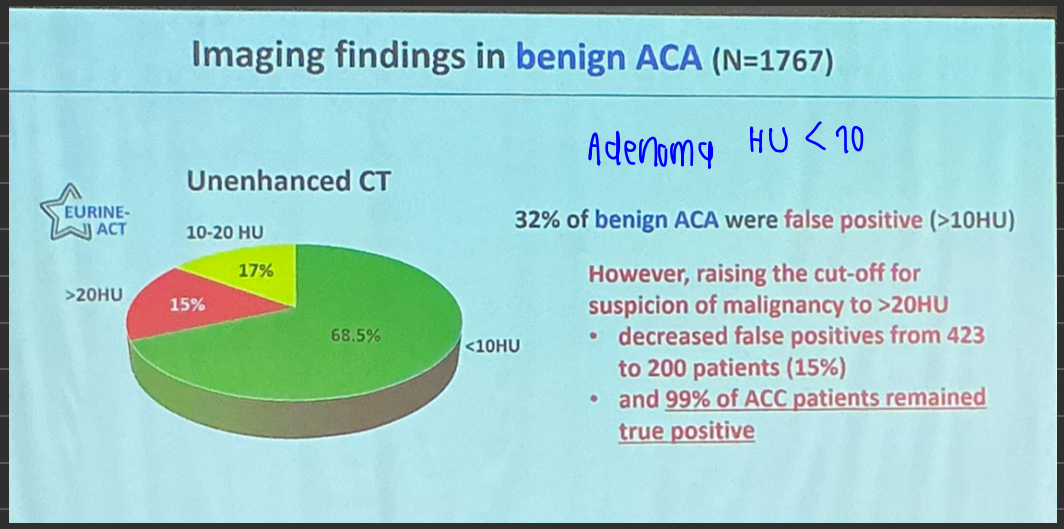


size ഉയരുമ്പോൾ ACC

HU കുറയും ACC.

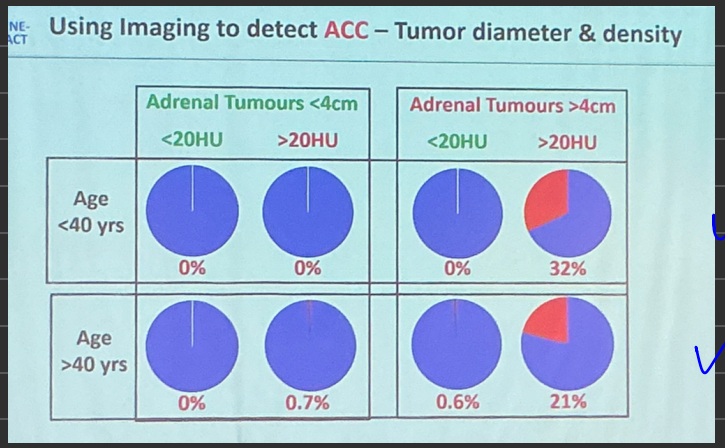
Adrenocortical carcinoma

size ഉയരുമ്പോൾ, HU കുറയും

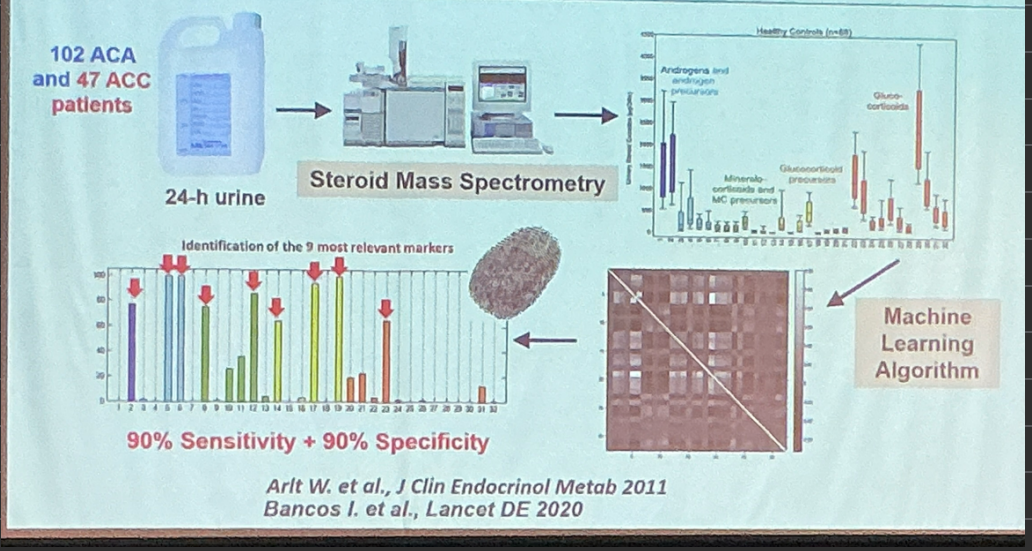


Adenoma HU < 10

tumor size, HU



Urine Steroid Metabolomics for the detection of adrenal cancer



Arlt W. et al., J Clin Endocrinol Metab 2011
Bancos I. et al., Lancet DE 2020

steroid metabomics for adrenal CA

EURINE-ACT

Learning Points – Adrenal Tumours

IMSR

- **Exclude Hormone excess**
  - Primary aldosteronism
  - Cushing's syndrome, MACS
  - Adrenal androgen and steroid precursor excess
  - Phaeochromocytoma
- **Exclude ACC, be suspicious if:**
  - Tumour diameter >4cm AND
  - Tumour attenuation >20HU on unenhanced CT

  } USM
  - Mixed hormone excess (combination of cortisol, androgens, sometimes also oestradiol)
  - Be very alert in young patients (<40 years) with adrenal tumours

dia > 4cm , 20HU

urine steroid met.

ก้อน & อายุน้อย (!) *

A Case: Henry (70 years old)

- **Bladder Cancer**, transurethal resection

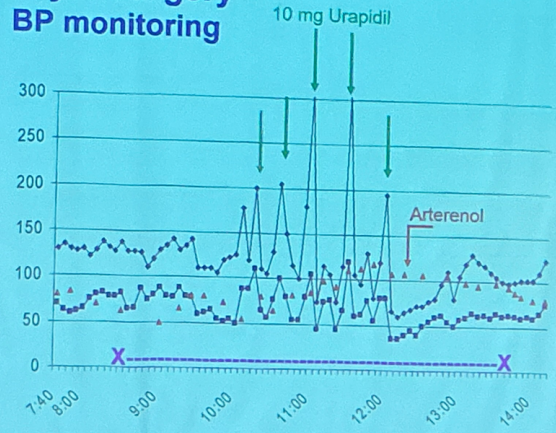
=> CT abdomen for follow-up at different hospital

=> Local recurrence of bladder tumour

- **„in addition, mass lesion (3.8 cm diameter) within the right adrenal gland, density in keeping with adrenal metastasis“**

=> Surgery for cystoprostatectomy and removal of adrenal mass

Henry's Surgery – BP monitoring

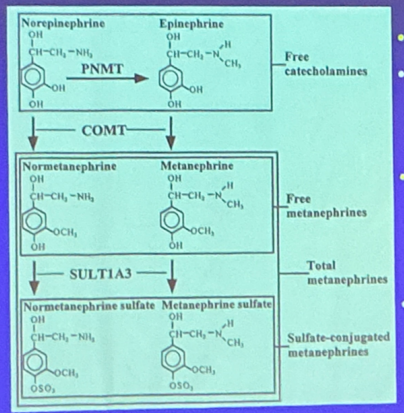


BP fluctuation ↑

Henry's surgery BP shoot.

α-blockade (phenoxybenzamine)

Phaeochromocytoma



Diagnosis

- **Most sensitive:**
- Plasma metanephrines
- **Slightly less sensitive, but slightly more specific:**
- 24-h urinary metanephrines

do metanephrines

Treatment

α-blockade (phenoxybenzamine, doxazosin

Surgery with special anaesthesia requirements

Ensure fluid resuscitation before, during and after surgery

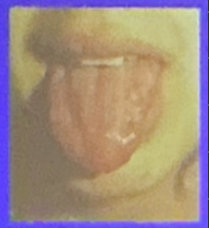
Phaeochromocytoma Genes

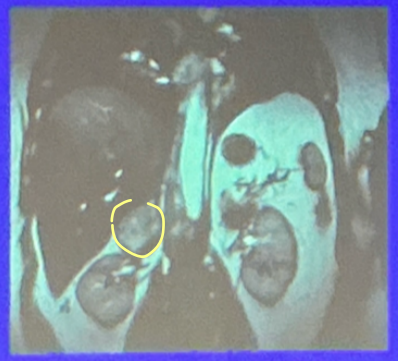
RET protooncogene

=> Multiple Endocrine Neoplasia (MEN) type 2 ☆

1) Phaeochromocytoma
2) Medullary thyroid carcinoma
3) parathyroid hyperplasia/tumour (MEN2A)

or mucocutaneous neuroma (MEN2B)





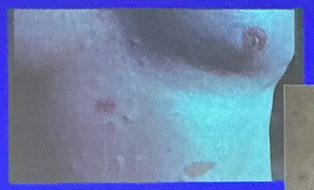
MTC

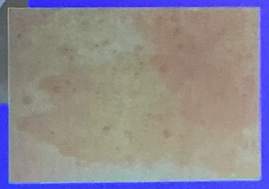
MEN 2 — RET protooncogene

Mucocutaneous lesion

Phaeochromocytoma Genes

NF-1 => Neurofibromatosis type 1 (von Recklinghausen)

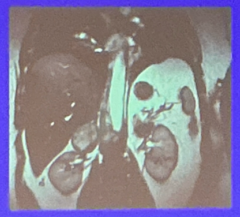


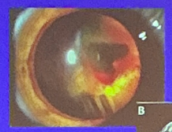


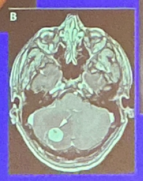
Phaeochromocytoma Genes

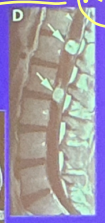
VHL => Von Hippel Lindau Syndrome

1) Haemangioblastoma (cerebellum, retina, spinal chord)
2) Phaeochromocytoma, renal angioma, renal cancer RCC ☆





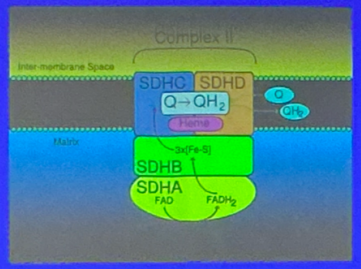




Phaeochromocytoma Genes

Succinyl Dehydrogenase Complex Subunit B and D

SDHC, SDHD => familial head and neck paraganglioma



SDHB

⇒ familial phaeochromocytoma

⇒ extraadrenal para-ganglioma (abdomen/pelvis, chest)

⇒ frequently malignant!